

Bridging What Futures Miss

Factor-Conditioned Moment-Matched Bridges for Commodity Portfolio Value at Risk

Felix Kwok

August 2026

Abstract

Historical simulation for a commodity cash-month position is often forced to use liquid futures as a proxy for an illiquid, short, or incompletely observed cash history. That substitution can omit the location, delivery-block, and prompt-period basis movements that dominate physical exposure near delivery. The objective is not merely to increase quoted VaR. It is to construct missing cash scenarios while retaining the empirical target-factor history and the calibrated linear dependence structure that governs joint portfolio loss. The proposed factor-conditioned moment-matched bridge creates residual scenarios satisfying exact finite-sample covariance and orthogonality constraints and imposes observed cash-level endpoints through a discrete bridge. In 115 gas monthly vintages, the complete 252-row output has median mapped-factor, cash-correlation, and residual-correlation RMSE of 0.0128, 0.0050, and 0.0058. The corresponding residual-correlation RMSE is 0.2570 for independently paired basis and 0.2617 for rolling cash. The bridge has the lowest error in every gas vintage; within the generated bridge segment, target errors are at numerical precision. These are construction-fidelity results, not an assertion that estimated second moments determine the future tail or preserve the complete higher-order residual distribution. A one-factor gas case and multi-factor power extension are also evaluated through leakage-free 95 percent VaR backtests from January 2017 through July 2026. Futures-only exception rates are 28.2 to 43.4 percent, while cash-aware methods produce 5.1 to 6.6 percent. The bridge does not uniformly dominate those methods in quantile loss. Its primary contribution is therefore disciplined preservation of relevant joint risk structure, with VaR magnitude a consequence rather than the design objective.

JEL classification: C14, C53, G17, Q40

1 Introduction

A historical Value-at-Risk model is only as relevant as the shocks applied to the position being measured. For a monthly natural-gas or power position, the economic exposure may be a physical cash index, a location-specific settlement, or an hourly delivery block. The longest and most liquid history, however, often belongs to a benchmark futures contract. Replacing the cash risk factor with that future is operationally attractive and economically incomplete. The replacement suppresses the basis between the traded hedge and the position that ultimately settles.

This distinction is particularly important in the cash month. Location constraints, weather, outages, fuel availability, day-ahead versus real-time market design, and the changing scarcity of transport or generation can move a cash index far more than its mapped monthly future. Supervisory guidance has long identified product, location, grade, and delivery-date mismatches as distinct commodity basis risks (Board of Governors of the Federal Reserve System 2009). Yet a risk manager who asks for one year of daily shocks before a future delivery month faces a practical obstacle: the desired cash month has not happened. The history required for a conventional historical simulation does not exist in calendar form.

The proposed solution is a *factor-conditioned moment-matched bridge*. It starts with the observed history of the liquid futures factors for the target contract, calibrates how cash changes co-move with those factors in comparable prompt windows, generates cross-sectionally correlated residual changes, and conditions the resulting path on observed cash endpoints. The endpoint adjustment has the same algebra as a discrete Brownian bridge. It does not assume that commodity prices follow continuous Brownian motion, nor does it infer a risk-neutral price. It is a finite-sample scenario transformation for historical VaR.

A larger VaR is not, by itself, evidence of a better model. Any sufficiently dispersed scenario set can reduce exceptions or worst-loss-to-VaR ratios. The scientific question is whether the dispersion arises from the economic structure of the exposure. For a cash position and its hedge factor, that requires the observed target-factor history, the calibrated cash-factor response, and the cross-location residual dependence to appear together in one synchronized scenario matrix. Portfolio risk then follows from that structure rather than from an arbitrary volatility addition.

The central question is not whether the constructed columns look plausible in isolation. It is whether a scenario matrix preserves the original cash-to-factor and cash-to-cash dependence relevant to portfolio loss while remaining point-in-time valid. The construction therefore imposes three exact sample conditions on the residual matrix: zero mean, orthogonality to the factor matrix, and a feasible target residual covariance. A final constant shift by series meets the cash endpoint without changing any centered covariance or correlation. This directly addresses factor double counting: systematic variation enters once, through the mapped futures factors, while residual variation is explicitly orthogonal to those factors.

The paper makes six contributions.

1. It formulates the missing cash-month history problem as conditional scenario generation rather than as an unconditional extrapolation of spot prices.
2. It gives a one-factor gas construction and a mapped multi-factor power construction in a common matrix notation.
3. It states the finite-sample invariants and feasibility conditions, including the role of positive-semidefinite projection when independently estimated moments are incompatible.
4. It measures structural fidelity across 115 gas vintages using mapped-factor, cash-correlation, and residual-correlation matrix errors, including dependence-aware bootstrap intervals.
5. It compares the proposal with futures-only shocks, an exact empirical convolution of futures and independently paired basis changes, and a rolling historical cash proxy in leakage-free monthly VaR backtests.
6. It examines Winter Storm Fern in January 2026 and separates supported risk-budget counterfactuals from unsupported claims about hedge-fund survival.

The structural result is stronger than the VaR ranking. In the complete gas scenario output, median residual-correlation RMSE is 0.0058 for the bridge, compared with 0.2570 for independent basis and 0.2617 for rolling cash. The bridge has lower residual-, cash-, and factor-structure error in all 115 gas vintages. Across six test portfolios, average bridge VaR is 9.6 to 21.8 times futures-only VaR, and futures-only exception rates range from 28.2 to 43.4 percent against a nominal 5 percent. Yet rolling cash and independent basis are competitive, and sometimes superior, in quantile loss. This distinction is central: the bridge demonstrably follows its calibrated joint risk structure; whether that estimated structure is the best forecast of a particular tail quantile is a separate empirical question.

2 Motivation and Related Literature

2.1 Why futures can miss cash-month risk

Commodity futures summarize expectations and risk premia for delivery over a specified contract period. Cash exposures retain additional dimensions. In gas, pipeline constraints and regional weather produce location basis. In power, non-storability, transmission congestion, unit outages, and hourly load create large differences across nodes, zones, market stages, and delivery blocks. Classical commodity models emphasize mean reversion, convenience yield, and term structure (Schwartz 1997). Electricity research adds pronounced seasonality, spikes, and delivery-period structure (Lucia and Schwartz 2002). Those features make a generic monthly future a useful factor but an unreliable complete proxy.

The public specifications illustrate the distinction. The ICE Henry LD1 contract (H) is a monthly cash-settled future based on the NYMEX Henry Hub settlement, with a 2,500 MMBtu contract size and a last trading day three business days before the contract period (Intercontinental Exchange 2026a). The PJM Western Hub real-time peak contract (PMI) settles to the average of published Western Hub real-time LMPs for hours ending 0800 through 2300 Eastern Prevailing Time on Monday through Friday, excluding NERC holidays (Intercontinental Exchange 2026b). A position in PJM PPL Zone day-ahead off-peak power is exposed to a different location, market stage, and block even when regional factors move together.

The basis omitted by futures-only VaR is not a statistical nuisance. It is the economic residual left after the hedge factor moves. If cash change for series i is $\Delta S_{i,t}$ and its mapped futures change is $\Delta F_{m(i),t}$, then the simple basis change is

$$\Delta b_{i,t} = \Delta S_{i,t} - \Delta F_{m(i),t}. \quad (1)$$

Treating $\Delta b_{i,t}$ as zero makes the hedge perfect by assumption. Sampling it independently preserves a marginal basis distribution but loses any factor-basis dependence. Reusing rolling cash changes preserves observed joint moves but may represent the wrong contract state, season, or delivery block for the future position month. The proposed bridge is designed between these two shortcuts.

2.2 Historical simulation and forecast evaluation

Historical simulation is attractive because it retains empirical skewness, heavy tails, and nonlinear cross-sectional events without specifying a Gaussian return law. It is also slow to adapt to regime shifts and can be asymmetric in how new shocks enter and leave a fixed window (Pritsker 2006). Weighted and filtered variants seek to combine empirical scenarios with current volatility information (Boudoukh et al. 1998; Barone-Adesi et al. 1999). The present paper addresses a different but related problem: the relevant cash series is partly missing by construction, so the scenario history itself must be synthesized before a historical quantile can be taken.

Moment-matching scenario generation has a substantial stochastic-programming literature. Algorithms can construct discrete outcomes that reproduce target moments and correlations (Høyland et al. 2003). Here, the constraints are narrower and more structural. Only the first two centered moments are imposed, residuals must be orthogonal to observed factors, and the factors themselves are not simulated. They are the actual 252 historical changes of the target monthly contracts. This keeps the empirical factor marginal intact.

Backtesting must examine both the frequency and temporal pattern of losses. The Kupiec test asks whether the exception rate equals the nominal tail probability (Kupiec 1995). Christoffersen's

framework additionally tests whether exceptions are independent through time (Christoffersen 1998). Quantile loss supplies a proper comparative score for the forecasted quantile, avoiding a ranking based only on a binary exception indicator (Gneiting 2011). Empirical studies of bank VaR have shown why statistical validation must accompany a model’s internal logic (Berkowitz and O’Brien 2002).

2.3 Relation to Brownian bridges

A continuous Brownian bridge is a stochastic process conditioned on endpoints. The construction below borrows only its discrete endpoint geometry. Given an unconditioned cumulative path U_k , the bridge subtracts the fraction k/n of the terminal miss and adds the same fraction of the desired endpoint. No claim is made that cash prices are diffusions, that increments are Gaussian, or that the result is a pricing measure. Calling the method a bridge describes an algebraic constraint, not a distributional assumption.

3 Problem Formulation

Let M be the position month and let $t = 1, \dots, n$ index the historical scenario dates available immediately before M . This study fixes $n = 252$. For p cash series, define the absolute daily change vector $\Delta \mathbf{S}_t \in \mathbb{R}^p$. Absolute changes are used instead of percentage returns because power prices can approach or cross zero and because linear commodity P&L is naturally expressed as quantity times \$/MWh or \$/MMBtu change.

For a fixed position vector $\mathbf{w}$, the scenario P&L is

$$\Pi_t = \mathbf{w}^\top \Delta \mathbf{S}_t. \quad (2)$$

The one-day 95 percent historical VaR is reported as a positive loss,

$$\text{VaR}_{0.95}(\mathbf{w}) = -Q_{0.05}(\Pi_1, \dots, \Pi_n), \quad (3)$$

where $Q_{0.05}$ is the empirical lower-tail order statistic. With 252 equally weighted scenarios, this is the 13th-worst P&L because $\lceil 0.05 \times 252 \rceil = 13$.

The forecasting information set is denoted $\mathcal{J}_{M-1}$ and contains only data known before the first day of M . Realized cash changes during M are reserved for evaluation. This distinction is essential. A method that uses the position month’s realized cash changes to construct its own scenarios is not a forecast and cannot support a point-in-time VaR decision.

Suppose k monthly futures factors are available. Let $\mathbf{F} \in \mathbb{R}^{n \times k}$ be their centered, standardized historical change matrix for the target contracts, and let $\mathbf{C}_F = \mathbf{F}^\top \mathbf{F} / (n-1)$ be its sample correlation matrix. Every cash series i is mapped to a factor $m(i)$. Gas is the transparent one-factor case; power uses several delivery-block and market-stage factors.

The objective is to construct a synthetic cash matrix $\tilde{\mathbf{S}} \in \mathbb{R}^{n \times p}$ that:

1. uses the observed factor paths in $\mathbf{F}$;
2. reproduces calibrated cash volatility and feasible correlation targets;
3. does not count factor variation again inside the residuals;
4. meets observed cash-level endpoints; and
5. contains exactly 252 complete, synchronized scenarios in $\mathcal{J}_{M-1}$.

4 Competing Scenario Methods

The empirical design compares four constructions. Each uses the same information cutoff, target month, portfolio, and 95 percent order-statistic rule.

4.1 Futures only

The futures-only method assigns each cash series its mapped factor change:

$$\Delta \tilde{S}_{i,t}^{\text{fut}} = \Delta F_{m(i),t}. \quad (4)$$

It preserves the target contract’s empirical factor history and assumes zero basis change. This is the most liquid and easiest benchmark, but it is exactly the proxy whose missing risk motivates the study.

4.2 Futures plus independently paired basis

Historical synchronized cash and factor observations before M produce 252 basis-change rows through Equation 1. The method forms portfolio factor P&L x_a and portfolio basis P&L y_b , then evaluates every sum $x_a + y_b$ for $a, b = 1, \dots, 252$. The resulting empirical product distribution contains 63,504 combinations.

This is an exact convolution of two empirical margins, not 63,504 independent market observations. It retains the complete cross-sectional structure within a factor row and within a basis row, while deliberately destroying their contemporaneous pairing. It is a useful benchmark for asking whether conditional dependence adds value beyond a realistic marginal basis distribution.

4.3 Rolling cash proxy

The rolling-cash method takes the latest 252 complete, synchronized cash changes available before M . It preserves every observed cash-to-cash relationship and tail event in that window. Its weakness is state mismatch: the rolling rows can mix seasons, prompt positions, changing mappings, and delivery calendars that do not correspond to the target contract’s factor history.

4.4 Factor-conditioned moment-matched bridge

The proposed method conditions cash changes on the observed target-factor matrix, generates residuals that satisfy exact finite-sample constraints, and then shifts increments to meet observed endpoints. Comparable prompt windows estimate the cash-factor loadings, cash volatilities, and cash cross-correlation. The most recent cash observation month is always earlier than M ; no position-month cash observation enters the forecast.

5 Mathematical Construction

5.1 Standardized factor decomposition

Let $\mathbf{Z}_S$ be a standardized cash-change matrix. Define a loading matrix $\mathbf{B} \in \mathbb{R}^{p \times k}$. In the reviewed mapping, each row has one nonzero entry: the estimated correlation ρ_i between cash series i and its mapped factor. Let

$$\mathbf{D} = \text{diag} \left(\sqrt{1 - \rho_1^2}, \dots, \sqrt{1 - \rho_p^2} \right). \quad (5)$$

The standardized cash construction is

$$\mathbf{Z}_S = \mathbf{F}\mathbf{B}^\top + \mathbf{E}\mathbf{D}, \quad (6)$$

where $\mathbf{E} \in \mathbb{R}^{n \times p}$ is the standardized residual matrix. For gas, $k = 1$ and $B_{i1} = \rho_i$. For power, distinct peak/off-peak and day-ahead/real-time contracts appear as columns of $\mathbf{F}$.

If $\mathbf{E}$ is orthogonal to $\mathbf{F}$ and has sample covariance $\mathbf{C}_\varepsilon$, Equation 6 implies

$$\mathbf{C}_S = \mathbf{B}\mathbf{C}_F\mathbf{B}^\top + \mathbf{D}\mathbf{C}_\varepsilon\mathbf{D}, \quad (7)$$

where $\mathbf{C}_S$ is the target standardized cash correlation. Solving for the raw residual target gives

$$\mathbf{C}_\varepsilon^{\text{raw}} = \mathbf{D}^{-1} (\mathbf{C}_S - \mathbf{B}\mathbf{C}_F\mathbf{B}^\top) \mathbf{D}^{-1}. \quad (8)$$

This equation prevents factor double counting. If residuals were sampled from the unconditional cash covariance and then added to the factor component, the factor-related covariance already present in cash would enter twice. Subtracting $\mathbf{B}\mathbf{C}_F\mathbf{B}^\top$ first isolates the conditional component.

5.2 Positive-semidefinite feasibility

Separately estimated loadings and correlation matrices need not make Equation 8 positive semidefinite. Sampling noise, short calibration windows, factor collinearity, and model misspecification can all produce a negative eigenvalue. A covariance target with a negative eigenvalue cannot be generated by any real residual matrix.

The method therefore eigendecomposes the symmetric raw target, clips negative eigenvalues, applies any rank limit required by the residual subspace, and renormalizes the diagonal to one. This produces a feasible correlation matrix $\mathbf{C}_\varepsilon^*$. The procedure is related to the nearest-correlation problem studied by Higham (2002), although the implementation here is the direct eigenvalue projection required by the finite-sample design. The reported PSD adjustment is the largest absolute entrywise change between the raw and projected residual correlations.

Projection changes the interpretation of exact preservation. When the raw target is feasible, the synthetic matrix reproduces the calibrated cash correlation. When projection is required, it reproduces the *projected implied cash target* exactly, not an impossible raw matrix. The adjustment must therefore be monitored as model-risk evidence rather than hidden as numerical housekeeping.

5.3 Exact residual moment constraints

The generated residual matrix satisfies

$$\mathbf{1}^\top \mathbf{E} = \mathbf{0}^\top, \quad (9)$$

$$\mathbf{F}^\top \mathbf{E} = \mathbf{0}, \quad (10)$$

$$\frac{1}{n-1} \mathbf{E}^\top \mathbf{E} = \mathbf{C}_\varepsilon^*. \quad (11)$$

Equation 9 fixes each residual column's sample mean at zero. Equation 10 makes residuals sample-

orthogonal to every factor. Equation 11 fixes residual covariance exactly. A random starting matrix is projected into the orthogonal complement of the design matrix $[\mathbf{1}, \mathbf{F}]$, whitened, and recolored to the target residual correlation. Randomness selects one admissible orientation; it does not relax the constraints.

Let $r = \text{rank}([\mathbf{1}, \mathbf{F}])$. Feasibility requires

$$\text{rank}(\mathbf{C}_\varepsilon^*) \leq n - r. \quad (12)$$

With 252 rows and a small number of mapped factors, this condition usually leaves ample residual dimension. It becomes relevant when the number of factors or cash series is large, or when missing data sharply reduces n .

5.4 Scaling to cash changes

Let μ_S and σ_S be calibrated absolute cash change means and standard deviations. The unconditioned change matrix is

$$\mathbf{U} = \mathbf{1}\mu_S^\top + \mathbf{Z}_S \text{diag}(\sigma_S). \quad (13)$$

The factor columns remain the actual standardized target-contract history; therefore the method does not impose a Gaussian factor marginal. The generated residual orientation is second-moment matched and may be regarded as a controlled scenario completion around that empirical factor path.

5.5 Discrete endpoint conditioning

Let $\mathbf{d} = \mathbf{L}_R - \mathbf{L}_L$ be the required cash-level change between observed left and right endpoints. If $\mathbf{U}_k$ is the cumulative sum of unconditioned increments through row k , define

$$\mathbf{B}_k = \mathbf{U}_k - \frac{k}{n}\mathbf{U}_n + \frac{k}{n}\mathbf{d}, \quad k = 1, \dots, n. \quad (14)$$

The bridged increment in every row is equivalently

$$\Delta \mathbf{B}_t = \Delta \mathbf{U}_t + \frac{\mathbf{d} - \mathbf{U}_n}{n}. \quad (15)$$

Thus each series receives a constant shift across the gap. The shift enforces $\sum_t \Delta \mathbf{B}_t = \mathbf{d}$ while leaving all centered columns unchanged.

Proposition 1 (Endpoint invariance). *For any finite matrix $\mathbf{U}$ and any endpoint vector $\mathbf{d}$, the increment transformation in Equation 15 preserves the sample covariance matrix of $\mathbf{U}$ exactly and enforces the desired column sums. If the factor columns are centered, it also preserves every cash-factor sample covariance and correlation.*

The proof follows immediately because subtracting each bridged column mean removes the constant endpoint shift. The endpoint condition changes only the column means required to connect levels. It does not distort volatility, cash-to-cash correlation, or cash-to-factor correlation. This is the principal mathematical advantage over a row-specific ad hoc correction.

6 Empirical Design

6.1 Structural claims and forecast hypotheses

The study separates deterministic construction claims from empirical forecast claims. An equality imposed by an algorithm should be proved and numerically audited, not presented as a stochastic hypothesis with a p-value.

P1: Structural invariance. Within the generated bridge segment, the empirical target-factor path is retained and the feasible residual correlation, cash correlation, volatility, factor orthogonality, and endpoint constraints are satisfied to numerical precision. The complete 252-row output should remain materially closer to those targets than the competing constructions after real observed rows are stitched around the synthetic segment.

The following four hypotheses concern realized risk performance rather than mechanical equality.

H1: Omitted basis risk. Futures-only VaR will produce materially more than 5 percent cash P&L exceptions.

H2: Coverage improvement. The factor-conditioned bridge will move exception rates closer to 5 percent than futures-only VaR.

H3: Comparative accuracy. Conditioning and exact moment matching will reduce mean quantile loss relative to the two cash-aware benchmarks.

H4: Stress relevance. During Winter Storm Fern, bridge VaR will imply lower VaR-budgeted notional and smaller worst-loss-to-VaR ratios than futures-only VaR, without necessarily preventing all exceptions.

H1, H2, and H4 concern economic magnitude. H3 is a comparative forecast claim. P1 concerns whether the scenario generator follows its declared structure and must not be confused with out-of-sample calibration.

6.2 Sample and market coverage

The backtest covers position months from January 2017 through July 2026, a total of 115 monthly vintages for each commodity. The evaluation contains 2,473 gas business-day observations and 2,475 power business-day observations. Every forecast is formed before the first day of its position month and remains fixed for that month’s realized one-day cash P&L observations.

Underlying observations comprise licensed daily physical gas cash assessments, licensed hourly day-ahead and real-time power prices, public exchange contract definitions, and daily settlement histories for the mapped monthly contracts. No proprietary source name, database object, credential, or mapping table is required to understand the methodology. The licensed observations cannot be redistributed; the aggregate results and synthetic example below are suitable for external review.

The gas cross-section contains Henry Hub, Algonquin Citygate, Transco Zone 6 non-New York, Waha, and Chicago Citygate. The monthly factor is the publicly specified Henry LD1 (H) future. The power cross-section starts from four public cash indices: PJM Western Hub real-time, PJM PPL Zone day-ahead, NYISO New York City Zone day-ahead, and CAISO SP15 day-ahead. Each index is split into peak and off-peak blocks, producing eight cash series. The corresponding monthly peak/off-peak and day-ahead/real-time futures are selected by location and market stage. PJM Western Hub examples include PMI for real-time peak, OPJ for real-time off-peak, PJC for day-ahead peak, and PJD for day-ahead off-peak.

Hourly power observations are aggregated on the market’s local prevailing-time calendar. PJM and NYISO use 5 x 16 peak schedules; CAISO uses its 6 x 16 schedule. Off-peak is the complementary set of

valid hourly observations. Day-ahead prices are assigned to the quote day preceding physical delivery; real-time prices are assigned to the delivery day. This distinction prevents a one-day information shift from masquerading as cash-factor correlation. Actual 23- and 25-hour daylight-saving days are retained through the underlying hourly calendar rather than forced into a synthetic 24-hour day.

6.3 Point-in-time monthly vintages

For position month M , the information cutoff is the final calendar day before M . The latest realized power cash observation month is $M - 1$. Comparable left and right prompt windows pair factor contract months with cash observation months that are already complete. Gas windows follow the public prompt-expiry calendar; power windows follow calendar delivery blocks and distinguish day-ahead from real-time quote dates. Missing levels are forward-filled only within the available factor calendar before changes are taken.

Each method returns exactly 252 complete marginal scenarios. If a stitched bridge contains one fewer row because two calendars meet at an anchor, the immediately preceding complete realized cash change is used to fill the marginal sample. This does not import future information. For the independent-basis benchmark, the two 252-row margins form the exact 252 by 252 product distribution described earlier.

6.4 Test portfolios

Positions are normalized so that comparisons reflect scenario construction, not arbitrary gross size.

Commodity	Portfolio	Composition	Gross unit
Gas	Algonquin	100% Algonquin Citygate	1 MMBtu
Gas	Northeast	50% Algonquin, 50% Transco Zone 6 non-NY	1 MMBtu
Gas	Diversified	Equal weight across five locations	1 MMBtu
Power	PJM West RT peak	100% PJM Western Hub RT peak	1 MWh
Power	Concentrated PJM	Equal weight across PJM West RT and PPL DA, peak and off-peak	1 MWh
Power	Diversified	Equal weight across all eight location-block series	1 MWh

The portfolios are intentionally simple. They expose single-location, within-region, and cross-region behavior without fitting weights to the backtest. They are not presented as investable strategies.

6.5 Evaluation statistics

For observation t , let q_t be the forecasted 5 percent P&L quantile and let Y_t be realized cash P&L. The exception indicator is $I_t = \mathbb{1}\{Y_t < q_t\}$. The exception rate is $\hat{p} = T^{-1} \sum_t I_t$. The Kupiec likelihood-ratio statistic compares the null probability $\alpha = 0.05$ with $\hat{p}$:

$$LR_{UC} = -2 \log \left[\frac{\alpha^x (1 - \alpha)^{T-x}}{\hat{p}^x (1 - \hat{p})^{T-x}} \right], \quad (16)$$

where $x = \sum_t I_t$. Christoffersen's independence statistic compares a common transition probability with separate exception probabilities following exception and non-exception days. Conditional cover-

age is the sum of the unconditional and independence statistics and is evaluated against a chi-squared distribution with two degrees of freedom.

The quantile score is

$$L_\alpha(Y_t, q_t) = (\alpha - \mathbb{1}\{Y_t < q_t\})(Y_t - q_t). \quad (17)$$

Lower mean quantile loss is better. Exception severity is also recorded as the loss beyond q_t divided by $|q_t|$. The tests are descriptive rather than a multiple-testing exercise. Monthly forecast fixing and persistent commodity regimes can induce dependence; the independence results are therefore substantive diagnostics, not mere technical rejections.

6.6 Structural-fidelity statistics

Forecast coverage asks whether a quantile is calibrated. Structural fidelity asks a different question: whether a scenario construction retains the joint linear risk structure supplied to it. For gas vintage m , let ρ_m be the calibrated vector of cash-to-Henry-Hub correlations, $\mathbf{C}_{S,m}$ the feasible cash correlation matrix, and $\mathbf{C}_{\varepsilon,m}$ the historical standardized residual-correlation matrix after the Henry Hub component is removed. For method j , corresponding quantities are reconstructed from its complete 252-row scenario output.

Vector error is measured by

$$\text{RMSE}_{\rho,jm} = \left[\frac{1}{p} \sum_{i=1}^p (\hat{\rho}_{ijm} - \rho_{im})^2 \right]^{1/2}. \quad (18)$$

For a correlation matrix $\mathbf{A}$ and target $\mathbf{A}^*$, the reported off-diagonal error is

$$\text{RMSE}_{A,jm} = \left[\frac{1}{p(p-1)} \sum_{i \neq l} (A_{il,jm} - A_{il,m}^*)^2 \right]^{1/2}. \quad (19)$$

The independent-basis moments are evaluated analytically over its complete 252-by-252 product distribution. Rolling cash is paired with the historical factor observed in its own proxy window; it therefore has a defined historical cash-factor relationship but not the target-contract factor path. Futures only has no residual matrix. The bridge is evaluated twice: exact invariants are audited within the generated synthetic segment, while comparative errors use the complete 252-row output after observed rows are stitched around that segment.

Monthly error differences are paired by vintage. Twelve-month circular block bootstrap intervals with 10,000 replications retain serial dependence and annual seasonality in those differences (Politis and Romano 1992). These intervals quantify target-fidelity differences; they do not turn an estimated calibration target into the unknown future distribution.

7 Structural-Fidelity Results

The one-factor gas case makes the distinction among methods transparent. Table 1 reports median monthly errors over 115 vintages. The bridge alone simultaneously retains the target Henry Hub margin, keeps row-level factor conditioning, and reconstructs the calibrated residual dependence. The independent-basis benchmark retains both empirical margins but replaces their contemporaneous pairing with a product distribution. Rolling cash retains actual historical cash rows, but those rows

belong to a different contract state and factor path. Futures only has no residual risk to preserve.

Table 1. Median full-output gas structural errors, January 2017 to July 2026. Correlation errors use off-diagonal matrix entries. The futures-only construction has no residual component.

Method	Target factor margin	Target row pairing	Factor RMSE	Cash-corr. RMSE	Residual-corr. RMSE
Futures only	Yes	No residual	1.0121	0.6498	–
Independent basis	Yes	No: product	0.2046	0.2581	0.2570
Rolling cash	No	Historical proxy	0.1424	0.2681	0.2617
Bridge	Yes	Yes	0.0128	0.0050	0.0058

The bridge has lower factor-loading, cash-correlation, and residual-correlation error than every applicable benchmark in all 115 gas vintages. Relative to independent basis, the 95 percent block-bootstrap intervals for the mean bridge advantage are $[0.1842, 0.2611]$ for factor loading, $[0.2263, 0.3058]$ for cash correlation, and $[0.2225, 0.3029]$ for residual correlation. Relative to rolling cash, the corresponding intervals are $[0.1259, 0.1441]$, $[0.2309, 0.3157]$, and $[0.2278, 0.3134]$. Every interval is strictly positive.

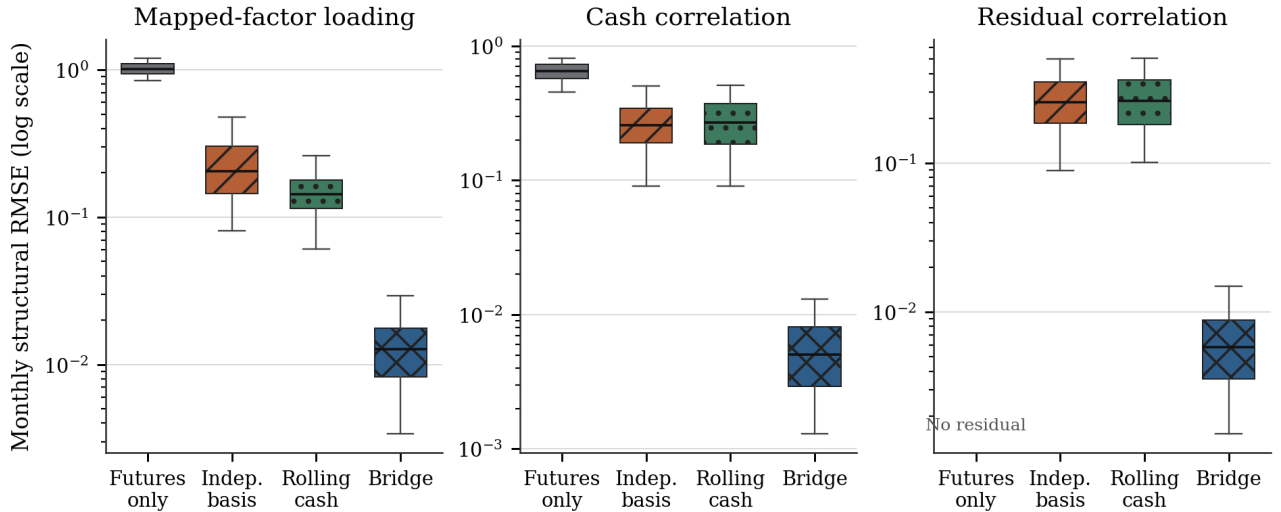


Figure 1. Gas structural-fidelity errors across 115 monthly vintages. Boxes show the 5th, 25th, 50th, 75th, and 95th percentiles on logarithmic axes. Metrics compare each complete 252-row output with the same calibrated target. Futures only has no residual matrix.

The complete-output errors in Figure 1 are deliberately not reported as numerical zero. The final VaR matrix contains real observed changes on both sides of the generated interval and, in a small number of cases, one preceding observed row needed to retain exactly 252 scenarios. Inside the synthetic bridge segment itself, the maximum gas mapped-factor error is 4.72×10^{-16} , cash-correlation error is 2.16×10^{-15} , volatility error is 1.78×10^{-14} , and endpoint error is 5.11×10^{-14} . The maximum gas PSD adjustment is 1.78×10^{-15} , confirming that the historical residual target is feasible up to floating-point precision.

This analysis demonstrates fidelity to a calibrated *linear* risk structure. It does not claim that the generated residual values retain the historical residual ranks, skewness, kurtosis, or tail copula. The target Henry Hub changes remain the actual empirical history; the higher-order residual distribution remains a separate model-risk dimension.

8 Backtest Results

8.1 Unconditional coverage

Figure 2 shows the forecast-calibration consequence of adding cash risk, not the bridge’s defining result. Futures-only VaR does not miss the nominal rate by a few observations. It records exceptions on 28.2 to 43.4 percent of realized cash days. The cash-aware methods compress this range to 5.1 to 6.6 percent. H1 is strongly supported, and H2 is supported economically across all portfolios.

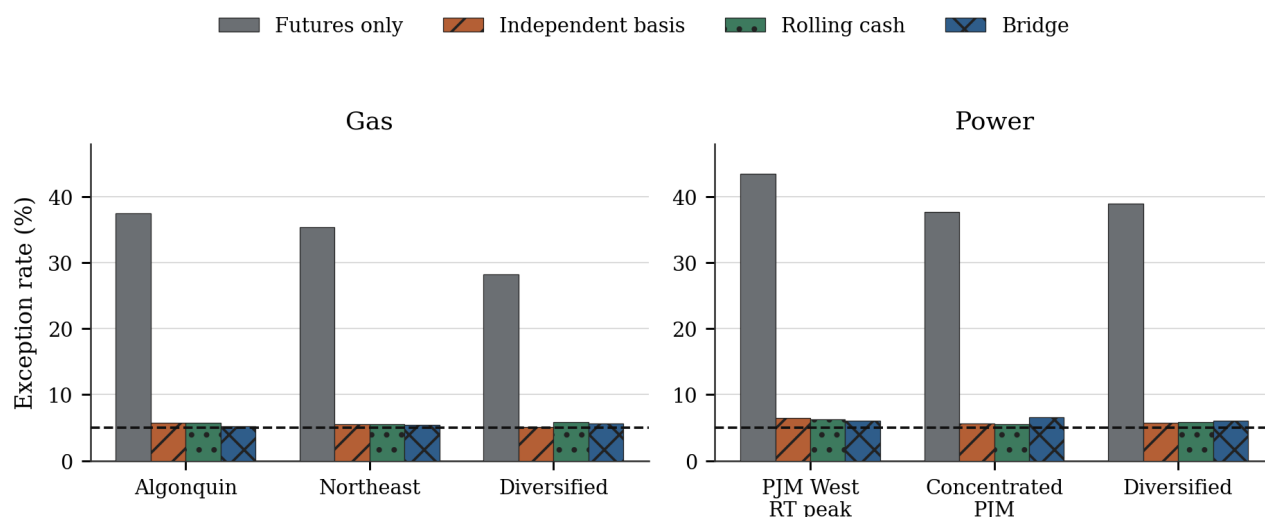


Figure 2. Exception rates by method and portfolio. The horizontal line is the nominal 5 percent tail probability. Forecasts are fixed before each position month and tested on realized one-day cash P&L.

Table 2 reports gas results. The bridge produces the lowest quantile loss for Algonquin and the concentrated Northeast portfolio, improving on rolling cash by 13.9 and 11.3 percent, respectively. Its unconditional coverage p-values are 0.561, 0.302, and 0.139, so the 5 percent null is not rejected for any gas portfolio. The diversified bridge is slightly worse than the independent and rolling benchmarks in quantile loss.

Table 2. Natural-gas backtest, January 2017 to July 2026. VaR and quantile loss (QL) are dollars per normalized MMBtu. Each method has 2,473 realized observations.

Portfolio	Method	Mean VaR	Exceptions	Rate	Mean QL	Kupiec p
Algonquin	Futures only	0.144	926	37.44%	0.491	<0.001
Algonquin	Independent basis	2.627	143	5.78%	0.357	0.081
Algonquin	Rolling cash	2.587	142	5.74%	0.355	0.098
Algonquin	Bridge	3.152	130	5.26%	0.306	0.561
Northeast	Futures only	0.144	873	35.30%	0.372	<0.001
Northeast	Independent basis	1.833	137	5.54%	0.287	0.226
Northeast	Rolling cash	1.796	137	5.54%	0.286	0.226
Northeast	Bridge	2.479	135	5.46%	0.253	0.302
Diversified	Futures only	0.144	697	28.18%	0.191	<0.001
Diversified	Independent basis	0.897	127	5.14%	0.154	0.758
Diversified	Rolling cash	0.819	145	5.86%	0.154	0.055

Portfolio	Method	Mean VaR	Exceptions	Rate	Mean QL	Kupiec p
Diversified	Bridge	1.444	140	5.66%	0.156	0.139

Power is less favorable to the bridge as a comparative forecast, as Table 3 shows. Bridge exception rates are 6.02, 6.63, and 6.06 percent. The corresponding Kupiec p-values of 0.024, less than 0.001, and 0.019 reject nominal coverage at 5 percent. The independent-basis and rolling-cash methods have lower quantile loss for Western Hub and concentrated PJM; the bridge is essentially tied with independent basis in the diversified portfolio. H3 is therefore not supported uniformly.

Table 3. Power backtest, January 2017 to July 2026. VaR and quantile loss are dollars per normalized MWh. Each method has 2,475 realized observations.

Portfolio	Method	Mean VaR	Exceptions	Rate	Mean QL	Kupiec p
PJM West RT peak	Futures only	1.672	1,074	43.39%	6.099	<0.001
PJM West RT peak	Independent basis	24.054	160	6.46%	3.317	0.001
PJM West RT peak	Rolling cash	24.091	154	6.22%	3.297	0.007
PJM West RT peak	Bridge	36.433	149	6.02%	3.750	0.024
Concentrated PJM	Futures only	1.374	933	37.70%	2.617	<0.001
Concentrated PJM	Independent basis	11.725	140	5.66%	1.580	0.142
Concentrated PJM	Rolling cash	11.486	138	5.58%	1.583	0.197
Concentrated PJM	Bridge	16.010	164	6.63%	1.625	<0.001
Diversified	Futures only	1.336	963	38.91%	2.079	<0.001
Diversified	Independent basis	9.847	141	5.70%	1.299	0.119
Diversified	Rolling cash	9.404	144	5.82%	1.302	0.068
Diversified	Bridge	12.886	150	6.06%	1.300	0.019

Figure 3 compares mean quantile loss directly. Futures-only is worst in every portfolio. Among cash-aware methods, there is no universal winner. The bridge’s additional structure matters in the concentrated gas cases; in power, the same structure can make the forecast more conservative without improving the quantile score.

8.2 VaR magnitude and position budgeting

Average bridge VaR is 21.8 times futures-only VaR for Algonquin and PJM Western Hub RT peak, 17.2 times for concentrated Northeast gas, 11.7 times for concentrated PJM power, 10.0 times for diversified gas, and 9.6 times for diversified power. These ratios do not imply that the bridge’s level is always optimal, nor do they validate the model merely because it is more conservative. They quantify how little of the calibrated cash dispersion is represented by the futures-only proxy. The structural results above establish where the additional dispersion comes from: mapped factor response and conditional residual dependence, rather than an arbitrary scalar volatility multiplier.

For a fixed per-day VaR budget K , a linear position’s admissible quantity is $Q = K/V$, where V is per-unit VaR. Holding everything else equal, the mean bridge estimates would reduce admissible

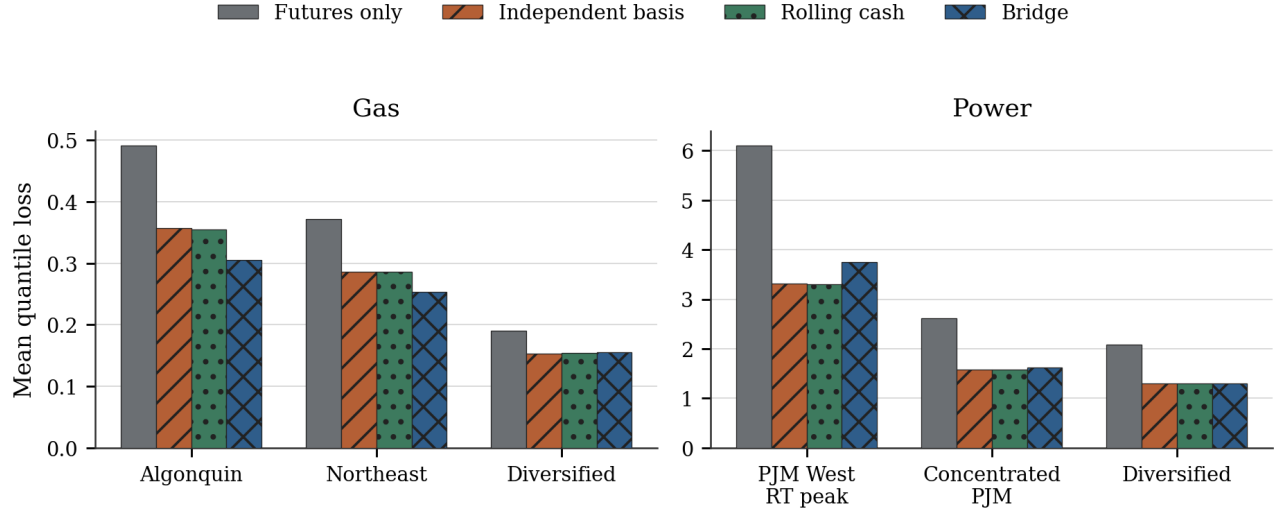


Figure 3. Mean 5 percent quantile loss. Lower is better; gas and power subplots have different dollar scales.

quantity to 4.6 to 10.4 percent of the futures-only quantity, a reduction of 89.6 to 95.4 percent. This calculation is a valid model-based risk-budget counterfactual. It is not evidence about the actual leverage, hedges, options, liquidity, or capital of any named market participant.

8.3 Independence and conditional coverage

Every method and portfolio rejects Christoffersen conditional coverage at conventional levels. The bridge improves exception frequency, but its exceptions remain clustered. For example, the Algonquin bridge passes unconditional coverage with $p = 0.561$ while rejecting independence with $p = 2.44 \times 10^{-8}$. The diversified power bridge has unconditional $p = 0.019$ and independence $p = 1.31 \times 10^{-7}$.

Clustering is consistent with persistent weather and infrastructure regimes. Once the market enters a scarcity state, consecutive cash moves can remain beyond a forecast calibrated mainly on ordinary days. A finite 252-day historical distribution also gives equal weight to regimes that may not be equally relevant today. The result echoes the broader warning that historical simulation can adapt slowly to changes in conditional risk (Pritsker 2006).

8.4 Finite-sample diagnostics

P1 is supported mechanically. Across all 230 monthly commodity vintages, the largest factor-correlation error is 7.22×10^{-16} , the largest cash-correlation error relative to the feasible target is 2.16×10^{-15} , the largest volatility error is 1.42×10^{-13} , and the largest endpoint error is 1.28×10^{-13} . These are floating-point tolerances rather than economic misses.

Gas requires no material PSD correction: the maximum entrywise adjustment is 1.78×10^{-15} . Power is different. Eleven of 115 vintages have an adjustment above 10^{-8} , seven exceed 0.01, and the maximum is 0.4057 for the February 2026 position month. The corresponding raw minimum residual eigenvalue is -0.6882. Figure 4 shows that most power vintages are feasible to machine precision, punctuated by economically meaningful adjustments.

The February result is not a failure of eigendecomposition. It says that the estimated mapped-factor loadings, factor correlations, and cash correlations cannot all be true simultaneously under the proposed linear decomposition. A large adjustment should trigger investigation of regime change,

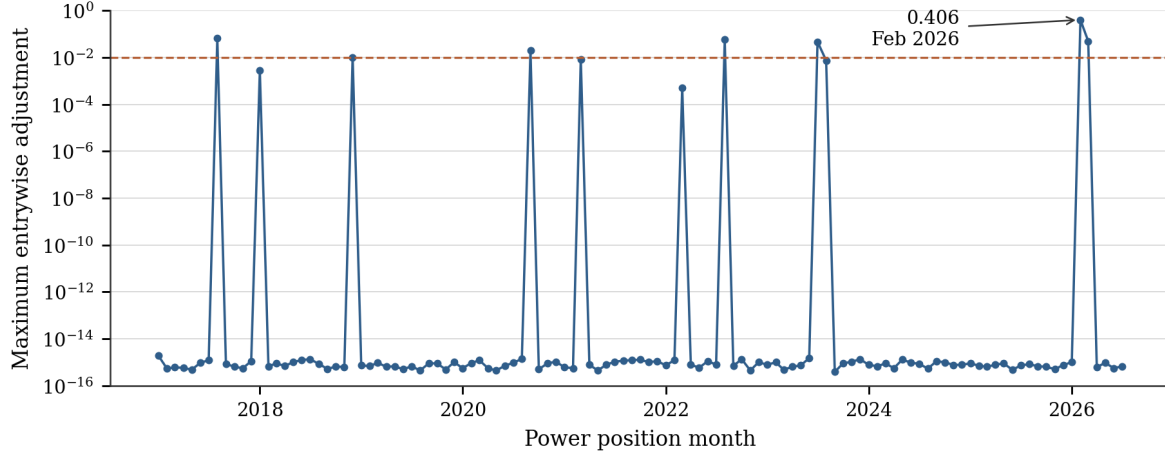


Figure 4. Maximum entrywise residual-correlation adjustment by power position month. The February 2026 value follows the January stress episode and is the sample maximum.

calibration window length, factor omission, and mapping stability. Exact generation of a projected target cannot repair a poor target.

9 Winter Storm Fern: January 2026

9.1 Event context

Winter Storm Fern provides a demanding out-of-sample episode near the end of the study. The U.S. Energy Information Administration reported a 360 Bcf Lower-48 storage withdrawal for the week ending January 30, 2026, the largest weekly net withdrawal in the history of its report (U.S. Energy Information Administration 2026). Natural-gas demand rose, production was constrained by freeze-offs, and regional cash prices moved violently.

PJM described the event as the strongest sustained cold since the 1990s. It reported loads above 130 GW for eight consecutive days, average generation outages of roughly 18 to 19 GW, and approximately \$798 million in out-of-market uplift from January 24 through February 1 (PJM Interconnection 2026). The U.S. Department of Energy issued an emergency order on January 25 authorizing specified PJM generation to run through operational limits during the emergency (U.S. Department of Energy 2026). These are conditions in which location, fuel, and market-stage basis should matter most.

The case-study window is January 23 through January 30 and contains six business-day cash observations. It is not a standalone statistical sample. Its purpose is to reveal economic severity and the consequences of a VaR budget formed before January began.

9.2 Case-study results

All methods experience exceptions. The bridge nevertheless changes the scale of the risk estimate materially. Table 4 reports the worst realized loss and its ratio to forecast VaR. The independent-basis ratios, omitted from the compact table, are similar to rolling cash: 12.84, 23.08, and 32.78 for the three gas portfolios, and 2.74, 6.91, and 9.63 for the power portfolios.

Table 4. Winter Storm Fern case study, January 23-30, 2026. Loss and VaR use the normalized portfolio units; L/V is worst realized loss divided by forecast VaR.

Portfolio	Worst loss	Worst date	Futures L/V	Rolling L/V	Bridge VaR	Bridge L/V
Gas:	39.90	Jan. 29	233.30	12.29	7.79	5.12
Algonquin						
Gas: Northeast	66.34	Jan. 28	387.94	22.53	8.72	7.61
Gas: diversified	40.67	Jan. 28	237.86	34.30	4.58	8.89
Power: PJM	133.21	Jan. 28	68.66	2.74	71.82	1.85
West RT peak						
Power:	157.09	Jan. 28	76.44	6.59	46.83	3.35
concentrated						
PJM						
Power:	158.17	Jan. 28	82.49	8.90	36.63	4.32
diversified						

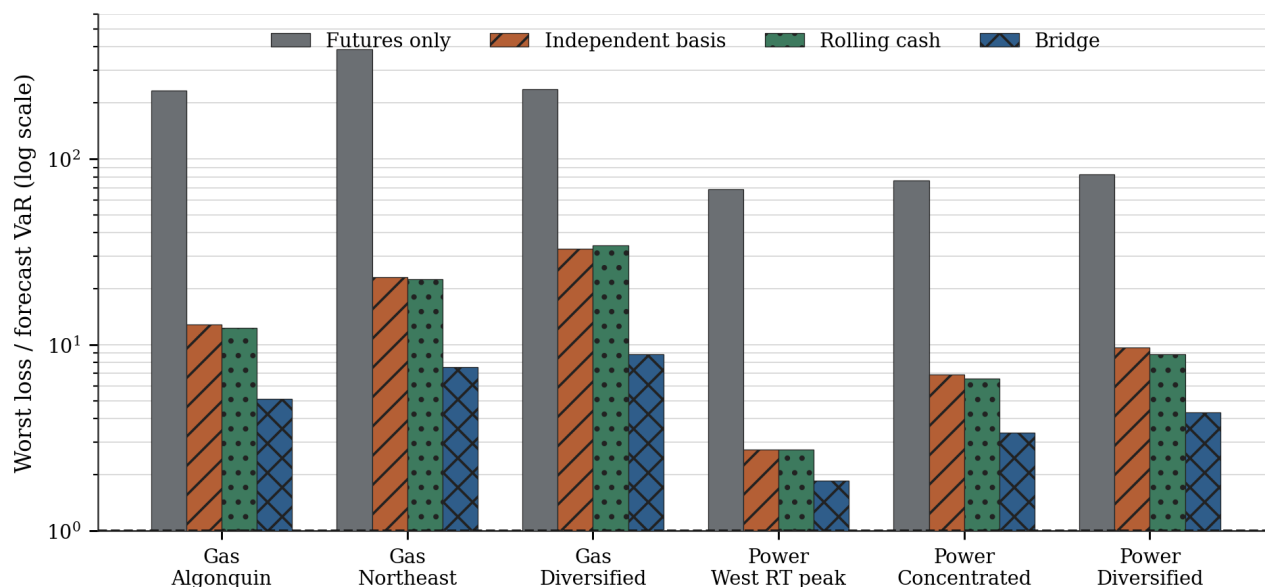


Figure 5. Worst realized loss divided by forecast VaR during the six-observation Fern window. The logarithmic axis is necessary because futures-only ratios reach nearly 388.

For Algonquin, the bridge raises January VaR from \$0.171/MMBtu to \$7.788/MMBtu, reducing the worst loss ratio from 233.3 to 5.1. For PJM Western Hub RT peak, it raises VaR from \$1.94/MWh to \$71.82/MWh, reducing the ratio from 68.7 to 1.85. Those are substantial improvements. They are also plainly insufficient to characterize the full event: the bridge records two or three exceptions in six observations, and several gas losses remain more than seven or eight VaRs.

9.3 What can and cannot be said about fund failures

The temptation to connect a large weather shock with public reports of fund closures should be resisted unless chronology and exposure are documented. Bloomberg reported on January 16, 2026 that Roscommon Analytics was closing its U.S. gas trading business after energy losses, including losses in U.S. power in the preceding year (Liao and Kumar 2026). That report predates the January 23-30 Fern window. It cannot be evidence that Fern caused the closure. Reuters subsequently reported that hedge

funds globally gained about 2.2 percent in January, while U.S. natural-gas futures rose 140 percent from January 20 through January 28 (Mackenzie 2026). Public performance was heterogeneous, not a documented wave of Fern-induced failures.

Amaranth Advisors is a relevant historical warning, not a Fern comparison. The CFTC reported that Amaranth lost approximately \$6 billion, about two-thirds of its value, in September 2006, largely from natural-gas positions on NYMEX and ICE (U.S. Commodity Futures Trading Commission 2007). It demonstrates how concentrated commodity spread exposure and leverage can overwhelm a fund. This paper has no position-level data with which to claim that the proposed bridge would have prevented that loss or any 2026 closure.

The defensible counterfactual is narrower. If a desk had enforced a fixed VaR budget using only mapped futures changes, the estimates in this study would have permitted much larger cash positions than the bridge. A bridge-based limit would therefore have forced material deleveraging before January. Whether that would have prevented insolvency depends on option convexity, intraday moves, margin, liquidity, hedge slippage, concentration, stop-loss governance, and available capital. None is observed here.

10 Interpretation and Model-Risk Implications

10.1 What the bridge corrects

The bridge corrects a specific structural omission: it restores conditional cash basis volatility and cross-sectional dependence to a target futures history. The gas fidelity results show that this restoration survives assembly of the complete VaR matrix and is not merely an internal algebraic assertion. The method is most useful when the desired cash month lacks a directly comparable one-year history, the mapped futures factor is observable, and historical prompt windows provide enough information to calibrate the cash-factor relationship.

The construction also makes model choices inspectable. Cash-factor correlations are explicit. Residual correlations are derived after subtracting the systematic factor component. Moment constraints are testable. Endpoint errors are measured. PSD adjustment is reported. These properties are preferable to an opaque synthetic path whose visual smoothness substitutes for statistical definition.

10.2 Why exact invariants do not guarantee exact coverage

The mathematical invariants are conditional on estimated targets. They say that the generated sample has the volatility and correlations requested by the model. They do not say that those targets equal next month's distribution. Four gaps remain.

First, volatility, correlation, and endpoints do not determine tail shape. Second, a linear factor decomposition can omit nonlinear scarcity behavior. Third, equal-weighted history treats stale and current regimes alike. Fourth, monthly calibration cannot anticipate every outage, weather forecast revision, or fuel-market discontinuity. Storm Fern makes all four visible.

The power results are especially instructive. The bridge is mechanically exact relative to its projected target and economically superior to futures only, yet it is not the best quantile forecast in several portfolios. More factors increase economic specificity but also increase estimation burden. Correlated peak, off-peak, day-ahead, and real-time factors can leave a residual target that is infeasible or unstable. The larger February 2026 PSD adjustment is evidence that the mapping structure should be challenged after a regime shock.

10.3 Why the simple cash-aware benchmarks remain competitive

Rolling cash requires no decomposition and therefore retains nonlinear cash events exactly as they occurred. Its weakness is relevance to the target month, not internal covariance. Independent basis sampling removes factor-basis timing, but the broad marginal basis distribution can still capture most of the 5 percent tail. Their strong performance means the bridge’s value should be judged against cash-aware alternatives, not only against an obviously incomplete futures proxy.

Similar exception rates do not imply similar risk structures. A conservative unconditional cash margin can place the 5 percent quantile adequately while misrepresenting how location residuals diversify, how cash co-moves with a hedge, or how a spread portfolio responds to the target contract. The structural audit therefore complements rather than duplicates the quantile backtest. It is especially relevant for portfolios containing several locations, partial futures hedges, or offsetting physical positions.

The comparison also clarifies that conditional correlation is not automatically valuable at every quantile and for every portfolio. If basis tails dominate and their factor timing is weak or unstable, preserving an estimated correlation may add conservatism without improving quantile loss. Conversely, in concentrated gas portfolios, the bridge’s dependence structure materially improves the score.

10.4 Governance use

A production governance framework should treat the bridge as one layer in a stack of controls:

1. **Primary historical VaR:** use the bridge where cash-month history is missing and mappings pass economic review.
2. **Benchmark challenge:** report futures-only, rolling-cash, and independent- basis estimates beside the bridge.
3. **PSD trigger:** escalate large residual-correlation adjustments and document whether the issue is noise, omitted factors, or a regime break.
4. **Exception monitoring:** track unconditional and conditional coverage by portfolio, month, and market regime.
5. **Stress and liquidity overlays:** add named weather, outage, congestion, pipeline, and margin scenarios that are not expected to appear once in a 252-day sample.
6. **Position-limit discipline:** translate per-unit VaR into quantity only after concentration, liquidity, option, and basis add-ons are applied.

The Federal Reserve’s observation that commodity VaR is complicated by location, delivery, event, and seasonal basis risk is directly relevant (Board of Governors of the Federal Reserve System 2009). A bridge can represent some of those dimensions. It cannot make a 95 percent quantile a complete solvency measure.

11 Limitations and Extensions

Several limitations define the current evidence.

Licensed data and replication. The empirical observations cannot be redistributed. Contract definitions are public, and the algorithm is fully specified, but an external replication must source equivalent cash, hourly, and settlement histories. The synthetic appendix permits exact verification of the algebra without licensed data.

Fixed test portfolios. Six normalized portfolios do not span the variety of physical books. Portfolios with opposite signs, options, transport, generation, or nonlinear settlement could rank

methods differently.

One confidence level and horizon. The study evaluates one-day 95 percent VaR because that is the stated operational use. Results do not establish performance at 99 percent, over multi-day liquidation horizons, or for expected shortfall.

Limited tail information. A 252-row marginal history identifies a 5 percent quantile from only 13 lower-tail rows. Even exact moment matching cannot create new empirical information about unprecedented regimes.

Static monthly forecasts. Forecasts are fixed for a month. Daily volatility rescaling or regime conditioning may respond faster, but can also introduce estimation noise and procyclicality.

Linear dependence. The cash-factor decomposition is linear and matches first and second moments. Tail dependence, threshold congestion, and asymmetric basis responses are not constrained.

Selection and multiple comparisons. Markets were selected to provide geographic and structural diversity, not randomly. Reported p-values are not adjusted for testing multiple portfolios.

Promising extensions follow directly. A regime-conditioned bridge could weight historical rows by temperature, load, storage, outage, or volatility state while retaining point-in-time controls. Robust or shrinkage covariance could reduce power PSD instability. Nonlinear residual models or copulas could target tail dependence, provided they preserve auditable factor attribution. Expected shortfall and stress-loss objectives could supplement the 5 percent quantile. Finally, portfolio-level validation should include signed physical positions, options, and realized hedge P&L rather than only normalized delta-one units.

12 Conclusion

Commodity cash-month VaR cannot be made accurate by relabeling a futures shock as a cash shock. The empirical evidence in this paper shows the consequence: a futures-only historical simulation produces 28 to 43 percent exceptions against a nominal 5 percent across the studied gas and power portfolios.

The factor-conditioned moment-matched bridge offers a mathematically disciplined correction. It uses observed target futures factors, removes their systematic component before constructing residual dependence, imposes exact residual moment constraints, and connects observed cash endpoints without changing centered volatility or correlation. Gas provides a transparent one-factor case. Power shows both the benefit and cost of a richer factor set.

The structural audit supports the method's primary purpose. In every gas vintage, the complete bridge output is closest to the calibrated mapped-factor, cash-correlation, and residual-correlation targets, while the generated segment satisfies them to numerical precision. The VaR backtest provides a separate and more mixed result. Bridge exception rates fall to 5.3 to 6.6 percent, but rolling cash and independent basis are competitive in power quantile loss, all exception sequences cluster, and Winter Storm Fern still generates losses several times bridge VaR. Large PSD adjustments identify periods in which the factor model itself is under strain.

The appropriate conclusion is therefore structural rather than merely conservative. The bridge does not claim merit because it produces a larger VaR. It earns its role by retaining the empirical target-factor history, separating systematic and residual variation without double counting, reconstructing the feasible linear dependence matrix, and meeting observed price endpoints. VaR is then calculated from those scenarios. The method should still sit beside cash-aware benchmarks, regime diagnostics, liquidity controls, and severe but plausible stress tests. What futures miss is real; what a bridge restores is important; and what remains outside a 95 percent historical quantile must still be governed.

Acknowledgments

The author thanks Shichang Liu and Christian Vargas for their thoughtful discussion and practical insight into commodity cash-market risk. All remaining errors and interpretations are the author's own.

13 Appendix A: Algorithmic Summary

The following pseudocode is vendor-neutral and applies to either commodity. Power uses a multi-column factor matrix; gas uses one factor.

```
INPUT:
  position month M
  252-row target-contract factor history F known before M
  completed historical cash and factor calibration windows
  observed left and right cash endpoint levels
  reviewed cash-to-factor mapping

1. Enforce point-in-time cutoff at the final day before M.
2. Align cash levels to each factor's valid quote calendar.
3. Aggregate hourly power into market-defined peak/off-peak blocks.
4. Compute absolute cash and factor changes on synchronized rows.
5. Estimate cash means, volatilities, mapped loadings B,
   factor correlation C_F, and cash correlation C_S.
6. Derive raw residual correlation:
   C_e_raw = D^{-1} (C_S - B C_F B') D^{-1}.
7. Project C_e_raw to a feasible PSD and rank-constrained C_e*;
   record the maximum entrywise adjustment.
8. Generate E in the orthogonal complement of [1, F] such that:
   1' E = 0,
   F' E = 0,
   E' E / (n - 1) = C_e*.
9. Form standardized cash Z_S = F B' + E D and scale to
   calibrated absolute cash changes U.
10. Apply the endpoint bridge to every cash series:
    bridge increment = U_t + (required endpoint change - sum(U)) / n.
11. Stitch observed and synthetic segments, retain exactly 252 complete rows,
    and compute portfolio P&L and the 13th-worst empirical quantile.
12. Validate scenario count, cutoff, PSD adjustment, moments,
    correlations, volatility, and endpoints before releasing VaR.

OUTPUT:
  252 synchronized daily cash-change scenarios by location/block
  plus validation diagnostics.
```

14 Appendix B: Four-Scenario Synthetic Example

This deliberately small example demonstrates the exact constraints without any licensed observation. Let the four-row standardized factor be

$$\mathbf{f} = \frac{1}{\sqrt{20/3}}(-3, -1, 1, 3)^T. \quad (20)$$

Choose two mutually orthogonal, zero-mean, unit-sample-variance vectors

$$\mathbf{q}_1 = \sqrt{3/4}(1, -1, -1, 1)^\top, \quad (21)$$

$$\mathbf{q}_2 = \sqrt{3/20}(1, -3, 3, -1)^\top. \quad (22)$$

Set residual A to $\mathbf{q}_1$ and residual B to $0.5\mathbf{q}_1 + \sqrt{0.75}\mathbf{q}_2$. Both are orthogonal to the factor; their sample correlation is 0.5. Let cash-factor correlations be 0.6 and 0.3, cash volatilities be 2 and 3, and desired cumulative endpoint changes be 4 and -2. The resulting rows are:

Factor z	Resid. A	Resid. B	Raw A	Raw B	Bridge A	Bridge B
-1.162	0.866	0.768	-0.009	1.153	0.991	0.653
-0.387	-0.866	-1.439	-1.850	-4.467	-0.850	-4.967
0.387	-0.866	0.573	-0.921	1.989	0.079	1.489
1.162	0.866	0.098	2.780	1.325	3.780	0.825

The raw columns sum to zero. The bridge adds 1.0 to each A row and -0.5 to each B row, so the bridge sums are exactly 4 and -2. Both raw and bridged sample standard deviations are 2 and 3. Cash-factor correlations are exactly 0.6 and 0.3. The cash-to-cash correlation is

$$(0.6)(0.3) + \sqrt{1 - 0.6^2}\sqrt{1 - 0.3^2}(0.5) = 0.561576. \quad (23)$$

The endpoint operation changes none of these centered moments. This example also shows why residual columns sum to zero: zero mean is an explicit moment-matching constraint. The final bridged cash changes need not sum to zero; they sum to the observed endpoint difference.

15 Appendix C: Proof of the Endpoint Invariance

Let $\mathbf{U}$ be an $n \times p$ increment matrix and let $\mathbf{c} = (\mathbf{d} - \mathbf{1}^\top \mathbf{U})/n$. The bridged matrix is $\mathbf{B} = \mathbf{U} + \mathbf{1}\mathbf{c}$. Its sample mean is

$$\bar{\mathbf{B}} = \bar{\mathbf{U}} + \mathbf{c}. \quad (24)$$

Therefore

$$\mathbf{B} - \mathbf{1}\bar{\mathbf{B}} = \mathbf{U} + \mathbf{1}\mathbf{c} - \mathbf{1}(\bar{\mathbf{U}} + \mathbf{c}) = \mathbf{U} - \mathbf{1}\bar{\mathbf{U}}. \quad (25)$$

The centered matrices are identical, so every sample covariance, variance, and correlation is identical. Also,

$$\mathbf{1}^\top \mathbf{B} = \mathbf{1}^\top \mathbf{U} + n\mathbf{c} = \mathbf{d}. \quad (26)$$

If a factor matrix is centered, adding $\mathbf{1}\mathbf{c}$ contributes zero to its cross-product with the factors. Cash-factor covariances are therefore unchanged as well.

16 Appendix D: Detailed Diagnostic Interpretation

The principal diagnostics answer distinct questions and should not be collapsed into one pass/fail flag.

Diagnostic	Question answered	Interpretation
Scenario count	Is the VaR marginal exactly 252 complete rows?	A comparability and order-statistic control.
Cutoff	Does every source row precede the position month?	A leakage control.
Factor error	Does synthetic cash retain the mapped factor correlation target?	A factor-attribution control.
Cash error	Does the synthetic cross-section match the feasible implied target?	A portfolio-dependence control.
Full-output factor RMSE	Does the assembled 252-row matrix retain calibrated factor loadings?	A target-factor fidelity comparison across methods.
Full-output cash-correlation RMSE	Does the assembled cash matrix retain the target cross-section?	A portfolio-diversification fidelity comparison.
Full-output residual-correlation RMSE	Is conditional dependence retained after removing the factor?	A direct test of the bridge's distinguishing structure.
Volatility error	Are calibrated absolute-change scales preserved?	A marginal second-moment control.
Endpoint error	Does the cumulative path meet observed cash levels?	A price-level consistency control.
PSD adjustment	Was the raw residual target mathematically feasible?	A model-risk and regime-stability signal.
Kupiec p-value	Is exception frequency compatible with 5 percent?	An unconditional forecast diagnostic.
Independence p-value	Are exceptions temporally independent?	A regime and clustering diagnostic.
Quantile loss	Which forecast places the 5 percent quantile more accurately?	A comparative forecast score.

Exact moment diagnostics validate construction, while coverage and quantile loss validate forecasting. The bridge can pass the first group and fail the second. That distinction is not a contradiction; it is the reason both groups are reported.

References

- Barone-Adesi, Giovanni, Kostas Giannopoulos, and Les Vosper. 1999. "VaR Without Correlations for Portfolios of Derivative Securities." *The Journal of Futures Markets* 19 (5): 583–602. [https://doi.org/10.1002/\(SICI\)1096-9934\(199908\)19:5%3C583::AID-FUT5%3E3.0.CO;2-S](https://doi.org/10.1002/(SICI)1096-9934(199908)19:5%3C583::AID-FUT5%3E3.0.CO;2-S).
- Berkowitz, Jeremy, and James O'Brien. 2002. "How Accurate Are Value-at-Risk Models at Commercial Banks?" *The Journal of Finance* 57 (3): 1093–111. <https://doi.org/10.1111/1540-6261.00455>.
- Board of Governors of the Federal Reserve System. 2009. *Application of the Market Risk Rule in Bank Holding Companies and State Member Banks*. Supervisory Letter SR 09-1. <https://www.federalreserve.gov/boarddocs/srletters/2009/SR0901.htm>.
- Boudoukh, Jacob, Matthew Richardson, and Robert F. Whitelaw. 1998. "The Best of Both Worlds: A Hybrid Approach to Calculating Value at Risk." *Risk* 11 (5): 64–67. <https://doi.org/10.2139/ssrn.51420>.
- Christoffersen, Peter F. 1998. "Evaluating Interval Forecasts." *International Economic Review* 39 (4):

- 841–62. <https://doi.org/10.2307/2527341>.
- Gneiting, Tilmann. 2011. “Making and Evaluating Point Forecasts.” *Journal of the American Statistical Association* 106 (494): 746–62. <https://doi.org/10.1198/jasa.2011.r10138>.
- Higham, Nicholas J. 2002. “Computing the Nearest Correlation Matrix: A Problem from Finance.” *IMA Journal of Numerical Analysis* 22 (3): 329–43. <https://doi.org/10.1093/imanum/22.3.329>.
- Høyland, Kjetil, Michal Kaut, and Stein W. Wallace. 2003. “A Heuristic for Moment-Matching Scenario Generation.” *Computational Optimization and Applications* 24 (2–3): 169–85. <https://doi.org/10.1023/A:1021853807313>.
- Intercontinental Exchange. 2026a. “Henry LD1 Fixed Price Future: Contract Specifications.” <https://www.ice.com/products/6590258>.
- Intercontinental Exchange. 2026b. “PJM Western Hub Real-Time Peak (1 MW) Fixed Price Future: Contract Specifications.” <https://www.ice.com/products/6590369>.
- Kupiec, Paul H. 1995. “Techniques for Verifying the Accuracy of Risk Measurement Models.” *The Journal of Derivatives* 3 (2): 73–84. <https://doi.org/10.3905/jod.1995.407942>.
- Liao, Ruth, and Nishant Kumar. 2026. “Hedge Fund Roscommon Analytics Shuttles U.S. Gas Desk.” Bloomberg News, January 16. <https://www.bloomberg.com/news/articles/2026-01-16/hedge-fund-roscommon-analytics-shuttles-us-gas-desk>.
- Lucia, Julio J., and Eduardo S. Schwartz. 2002. “Electricity Prices and Power Derivatives: Evidence from the Nordic Power Exchange.” *Review of Derivatives Research* 5 (1): 5–50. <https://doi.org/10.1023/A:1013846631785>.
- Mackenzie, Nell. 2026. “Hedge Funds Jump into Volatile January to Reap Returns.” Reuters, February 5. <https://www.investing.com/news/economy-news/factboxhedge-funds-jump-into-volatile-january-to-reap-returns-4488555>.
- PJM Interconnection. 2026. “PJM Reviews January Cold Weather Operations.” February 6. <https://insidelines.pjm.com/pjm-reviews-january-cold-weather-operations/>.
- Politis, Dimitris N., and Joseph P. Romano. 1992. “A Circular Block-Resampling Procedure for Stationary Data.” In *Exploring the Limits of Bootstrap*, edited by Raoul LePage and Lynne Billard. John Wiley & Sons.
- Pritsker, Matthew. 2006. “The Hidden Dangers of Historical Simulation.” *Journal of Banking & Finance* 30 (2): 561–82. <https://doi.org/10.1016/j.jbankfin.2005.04.013>.
- Schwartz, Eduardo S. 1997. “The Stochastic Behavior of Commodity Prices: Implications for Valuation and Hedging.” *The Journal of Finance* 52 (3): 923–73. <https://doi.org/10.1111/j.1540-6261.1997.tb02721.x>.

- U.S. Commodity Futures Trading Commission. 2007. *Amaranth: CFTC Chief Economist's Overview*. <https://www.cftc.gov/sites/default/files/idc/groups/public/@newsroom/documents/speechandtestimony/opalukken-26.pdf>.
- U.S. Department of Energy. 2026. "Federal Power Act Section 202(c): PJM Interconnection Order No. 202-26-02." January 25. <https://www.energy.gov/ceser/federal-power-act-section-202c-pjm-interconnection-pjm-order-no-202-26-02>.
- U.S. Energy Information Administration. 2026. "Record Natural Gas Stock Withdrawals During Week Ending January 30, 2026." February 5. <https://www.eia.gov/todayinenergy/detail.php?id=67124>.